

Customer Purchase Analysis & Insights

-From Data Cleaning to Power BI Visualizations

1.Business Problem:

Retail businesses collect large volumes of customer transaction data, but many struggle to convert this data into meaningful insights that improve sales and customer experience. The objective of this project is to analyze retail customer data to identify purchasing patterns, customer segments, and key factors influencing sales. By leveraging data analytics techniques, the goal is to generate actionable insights that help business improve customer retention, optimize product offerings, and increase overall revenue.

2.Project Objectives:

1. Analyze customer purchase behavior and transaction trends.
2. Identify different customer segments based on buying patterns.
3. Determine the most profitable products and categories.
4. Understand factors influencing customer retention and churn.
5. Provide data-driven recommendations to improve marketing strategies and sales performance.

3.Executive Summary:

This report presents a comprehensive end-to-end analysis of customer behaviour using a real-world CSV dataset. The project was executed across three platforms : Google Colab for data preprocessing using Pandas, PostgreSQL (via PgAdmin) for structured business queries, and Power BI for interactive visual dashboards. The objective is to uncover actionable patterns in customer purchasing behaviour, segment performance, and sales trends.

1. Dataset loaded from CSV and cleaned using Pandas. Nulls handled with median imputation, columns standardised to lowercase.
2. New derived features engineered: age groups, net revenue, time-based columns.
3. 10+ SQL queries executed on PostgreSQL to answer critical business questions.
4. Power BI dashboard built via PgAdmin live connection with KPI cards and segment filters.

4.Dataset Description:

The dataset is a CSV file containing customer transaction records capturing demographic details, purchase history, product categories, and sales amounts.

Column Name	Data Type	Description
Customer ID	Integer	Unique identifier for each customer
Age	Integer	Age of the customer
Gender	String	Gender of the customer (Male / Female)

Item Purchased	String	Name of the item purchased
Category	String	Product category (e.g. Clothing, Footwear, Accessories)
Purchase Amount (USD)	Integer	Transaction value in US dollars
Location	String	Customer's state or city of residence
Size	String	Size of the item purchased (S / M / L / XL)
Color	String	Color variant of the item purchased
Season	String	Season of purchase (Spring / Summer / Fall / Winter)
Review Rating	Float	Customer rating for the purchase (1.0 – 5.0); 37 null values
Subscription Status	String	Whether the customer has an active subscription (Yes / No)
Shipping Type	String	Shipping method chosen (e.g. Standard, Express, Free Shipping)
Discount Applied	String	Whether a discount was applied (Yes / No)
Promo Code Used	String	Whether a promo code was used at checkout (Yes / No)
Previous Purchases	Integer	Number of prior purchases made by the customer
Payment Method	String	Mode of payment (e.g. Credit Card, PayPal, Cash)
Frequency of Purchases	String	How often the customer shops (e.g. Weekly, Bi-Weekly, Monthly)

4.Tools & Technologies:

Tool / Platform	Purpose
Python (Pandas)	Data loading, cleaning, EDA
Google Colab	Cloud notebook environment
NumPy	Numerical operations
Matplotlib / Seaborn	EDA visualisation
Supabase (PostgreSQL)	Relational DB for SQL analysis
PgAdmin 4	PostgreSQL GUI + Power BI connector
Power BI Desktop	Interactive business dashboard

5.Data Preprocessing & Cleaning:

Raw data is rarely analysis-ready. So we start with data cleaning. But before that we will load the CSV dataset into Google Colab Notebook with Python as programming language.

5.1. Data Loading :

Importing the csv dataset using Python Pandas as `pd.read_csv(filename)`.

5.2. Data Inspection:

The dataset was initially explored to understand its structure and characteristics. This includes:

- Reviewing the first few rows

	Customer ID	Age	Gender	Item Purchased	Category	Purchase Amount (USD)	Location	Size	Color	Season	Review Rating	Subscription Status	Shipping Type	Discount Applied	Promo Code Used	Previous Purchases	Payment Method	Frequency of Purchases
0	1	55	Male	Blouse	Clothing	53	Kentucky	L	Gray	Winter	3.1	Yes	Express	Yes	Yes	14	Venmo	Fortnightly
1	2	19	Male	Sweater	Clothing	64	Maine	L	Maroon	Winter	3.1	Yes	Express	Yes	Yes	2	Cash	Fortnightly
2	3	50	Male	Jeans	Clothing	73	Massachusetts	S	Maroon	Spring	3.1	Yes	Free Shipping	Yes	Yes	23	Credit Card	Weekly
3	4	21	Male	Sandals	Footwear	90	Rhode Island	M	Maroon	Spring	3.5	Yes	Next Day Air	Yes	Yes	49	PayPal	Weekly
4	5	45	Male	Blouse	Clothing	49	Oregon	M	Turquoise	Spring	2.7	Yes	Free Shipping	Yes	Yes	31	PayPal	Annually

- Checking the number of rows and columns using `df.shape`
(3900, 18)

- Examining column names

```
Index(['Customer ID', 'Age', 'Gender', 'Item Purchased', 'Category',  
      'Purchase Amount (USD)', 'Location', 'Size', 'Color', 'Season',  
      'Review Rating', 'Subscription Status', 'Shipping Type',  
      'Discount Applied', 'Promo Code Used', 'Previous Purchases',  
      'Payment Method', 'Frequency of Purchases'],  
      dtype='object')
```

- Identifying data information including data types and null values of each variable using `df.info()`

```
RangeIndex: 3900 entries, 0 to 3899  
Data columns (total 18 columns):  
#   Column                                Non-Null Count  Dtype  
---  -  
0   Customer ID                          3900 non-null   int64  
1   Age                                  3900 non-null   int64  
2   Gender                              3900 non-null   object  
3   Item Purchased                       3900 non-null   object  
4   Category                            3900 non-null   object  
5   Purchase Amount (USD)                3900 non-null   int64  
6   Location                             3900 non-null   object  
7   Size                                 3900 non-null   object  
8   Color                                3900 non-null   object  
9   Season                               3900 non-null   object  
10  Review Rating                        3863 non-null   float64  
11  Subscription Status                  3900 non-null   object  
12  Shipping Type                       3900 non-null   object  
13  Discount Applied                    3900 non-null   object  
14  Promo Code Used                     3900 non-null   object  
15  Previous Purchases                   3900 non-null   int64  
16  Payment Method                      3900 non-null   object  
17  Frequency of Purchases               3900 non-null   object  
dtypes: float64(1), int64(4), object(13)  
memory usage: 548.6+ KB
```

- Descriptive statistics were generated to understand the distribution of numerical features and detect potential anomalies.

	Customer ID	Age	Purchase Amount (USD)	Review Rating	Previous Purchases
count	3900.000000	3900.000000	3900.000000	3863.000000	3900.000000
mean	1950.500000	44.068462	59.764359	3.750065	25.351538
std	1125.977353	15.207589	23.685392	0.716983	14.447125
min	1.000000	18.000000	20.000000	2.500000	1.000000
25%	975.750000	31.000000	39.000000	3.100000	13.000000
50%	1950.500000	44.000000	60.000000	3.800000	25.000000
75%	2925.250000	57.000000	81.000000	4.400000	38.000000
max	3900.000000	70.000000	100.000000	5.000000	50.000000

5.3. Handling Missing Values

The dataset contained missing values in the Review Rating column. Instead of removing these records, the missing values were imputed using the median review rating within each product category. This approach preserved the dataset size while maintaining category-level consistency.

```

RangeIndex: 3900 entries, 0 to 3899
Data columns (total 18 columns):
#   Column                                Non-Null Count  Dtype
---  -
0   Customer ID                          3900 non-null   int64
1   Age                                   3900 non-null   int64
2   Gender                               3900 non-null   object
3   Item Purchased                       3900 non-null   object
4   Category                             3900 non-null   object
5   Purchase Amount (USD)                3900 non-null   int64
6   Location                             3900 non-null   object
7   Size                                  3900 non-null   object
8   Color                                 3900 non-null   object
9   Season                               3900 non-null   object
10  Review Rating                        3900 non-null   float64
11  Subscription Status                 3900 non-null   object
12  Shipping Type                       3900 non-null   object
13  Discount Applied                    3900 non-null   object
14  Promo Code Used                     3900 non-null   object
15  Previous Purchases                  3900 non-null   int64
16  Payment Method                      3900 non-null   object
17  Frequency of Purchases              3900 non-null   object
dtypes: float64(1), int64(4), object(13)
memory usage: 548.6+ KB

```

5.4. Standardizing Column Names

To ensure consistency and ease of analysis, column names were standardized by:

- Converting all column names to lowercase
- Replacing spaces with underscores

- Removing unnecessary special characters

This improves readability and prevents errors during analysis and coding.

```
Index(['customer_id', 'age', 'gender', 'item_purchased', 'category',
      'purchase_amount(usd)', 'location', 'size', 'color', 'season',
      'review_rating', 'subscription_status', 'shipping_type',
      'discount_applied', 'promo_code_used', 'previous_purchases',
      'payment_method', 'frequency_of_purchases'],
      dtype='object')
```

5.5. Feature Engineering

5.5.1. Age Group Segmentation:

Customer ages were categorized into four groups using quantile-based binning:

- Young Adult
- Adult
- Middle-aged
- Senior

This segmentation helps analyze purchasing patterns across different age demographics.

5.5.2. Purchase Frequency Conversion:

The categorical variable frequency_of_purchases was converted into a numerical variable purchase_frequency_days. Each purchase frequency category was mapped to an approximate number of days between purchases (e.g., Weekly = 7 days, Monthly = 30 days). This transformation enables easier quantitative analysis.

5.6. Removing Redundant Variables

A comparison between discount_applied and promo_code_used revealed that both variables contained identical information. Since they represented the same concept, the promo_code_used column was removed to avoid redundancy and simplify the dataset.

5.7. Data Validation

After completing the cleaning and transformation steps, the dataset was rechecked to confirm that:

- Missing values were handled
- Data types were appropriate
- Newly created variables were correctly generated

These steps ensured the dataset was reliable and ready for exploratory data analysis and customer behavior insights.

6. Database Integration (Google Colab → Supabase)

After data cleaning and processing, the updated csv file has to be uploaded in Supabase for further querying using Postgre SQL.

The below are the steps for integrating Google Colab and Supabase.

- 6.1 Create Supabase Project: Set up a PostgreSQL database on Supabase.
- 6.2 Obtain Database Credentials: Username, Host, Port, Database, Password.

6.3 Install Python Library: Install sqlalchemy and psycopg2-binary to connect Python with PostgreSQL.

6.4 Create SQLAlchemy Engine: Use SQLAlchemy's create_engine() to establish a connection to Supabase. Provides a flexible interface for executing SQL queries and loading data into Python.

6.5 Execute SQL Queries: Retrieve customer and transaction data for analysis.

7. Data Analysis using SQL queries

After data cleaning and processing, the updated csv file has to be uploaded in Supabase for further querying using Postgre SQL.

Querying the database using SQL allowed us to extract meaningful insights by answering key business questions as below.

7.1. Total revenue generated by male vs. female customers?

gender	total_revenue
Female	75191
Male	157890

7.2. Which customers used a discount but still spent more than the average purchase amount?

customer_id	purchase_amount(usd)
2	64
3	73
4	90
7	85
9	97
839 rows	

7.3. Top 5 products with the highest average review rating?

item_purchased	average_review_rating
Gloves	3.86
Sandals	3.84
Boots	3.82
Hat	3.80
Skirt	3.78

7.4. Compare the average Purchase Amounts between Standard and Express Shipping.

shipping_type	average_purchase_amount
Express	60.48
Standard	58.46

7.5. Do subscribed customers spend more? Compare average spend and total revenue between subscribers and non-subscribers.

subscription_status	customer_count	avg_purchase_amount	total_revenue
No	2847	59.87	170436.00
Yes	1053	59.49	62645.00

7.6. 5 products that have the highest percentage of purchases with discounts applied?

item_purchased	discounted_percentage
Hat	50.00
Sneakers	49.66
Coat	49.07
Sweater	48.17
Pants	47.37

7.7. Segment customers into New, Returning, and Loyal based on their total number of previous purchases, and show the count of each segment.

segment	count
Loyal	3116
New	83
Returning	701

7.8. Top 3 most purchased products within each category

rank	category	item_purchased	purchase_count
1	Accessories	Jewelry	171
2	Accessories	Sunglasses	161
2	Accessories	Belt	161
3	Accessories	Scarf	157
1	Clothing	Blouse	171
1	Clothing	Pants	171
2	Clothing	Shirt	169
3	Clothing	Dress	166
1	Footwear	Sandals	160
2	Footwear	Shoes	150
3	Footwear	Sneakers	145
1	Outerwear	Jacket	163
2	Outerwear	Coat	161

7.9. Are customers who are repeat buyers (more than 5 previous purchases) also likely to subscribe?

subscription_status	count
Yes	958
No	2518

7.10. What is the revenue contribution of each age group?

age_group	revenue
young adult	62143
middle-aged	59197
adult	55978
senior	55763

8. Visualization Using Power BI

Supabase PostgreSQL uses secure authentication and SSL configuration. Therefore, the database connection was first verified using pgAdmin before connecting it to Microsoft Power BI for visualization.

Connecting Supabase PostgreSQL to pgAdmin.

Before building dashboards, the Supabase database was connected to pgAdmin to verify tables and database structure.

8.1 Obtain Database Credentials from Supabase: From the Supabase dashboard, copy the host, database name, port, username, and password of the PostgreSQL database.

8.2 Create Server in pgAdmin: Open pgAdmin → Right-click Servers → Select Register → Server.

8.3 Enter Connection Details: In the Connection tab, entered the Supabase PostgreSQL credentials:

- Host name/address
- Port
- Database name
- Username and password.

8.4 Enable SSL Configuration: Configure SSL mode if required to establish a secure connection with the Supabase PostgreSQL database.

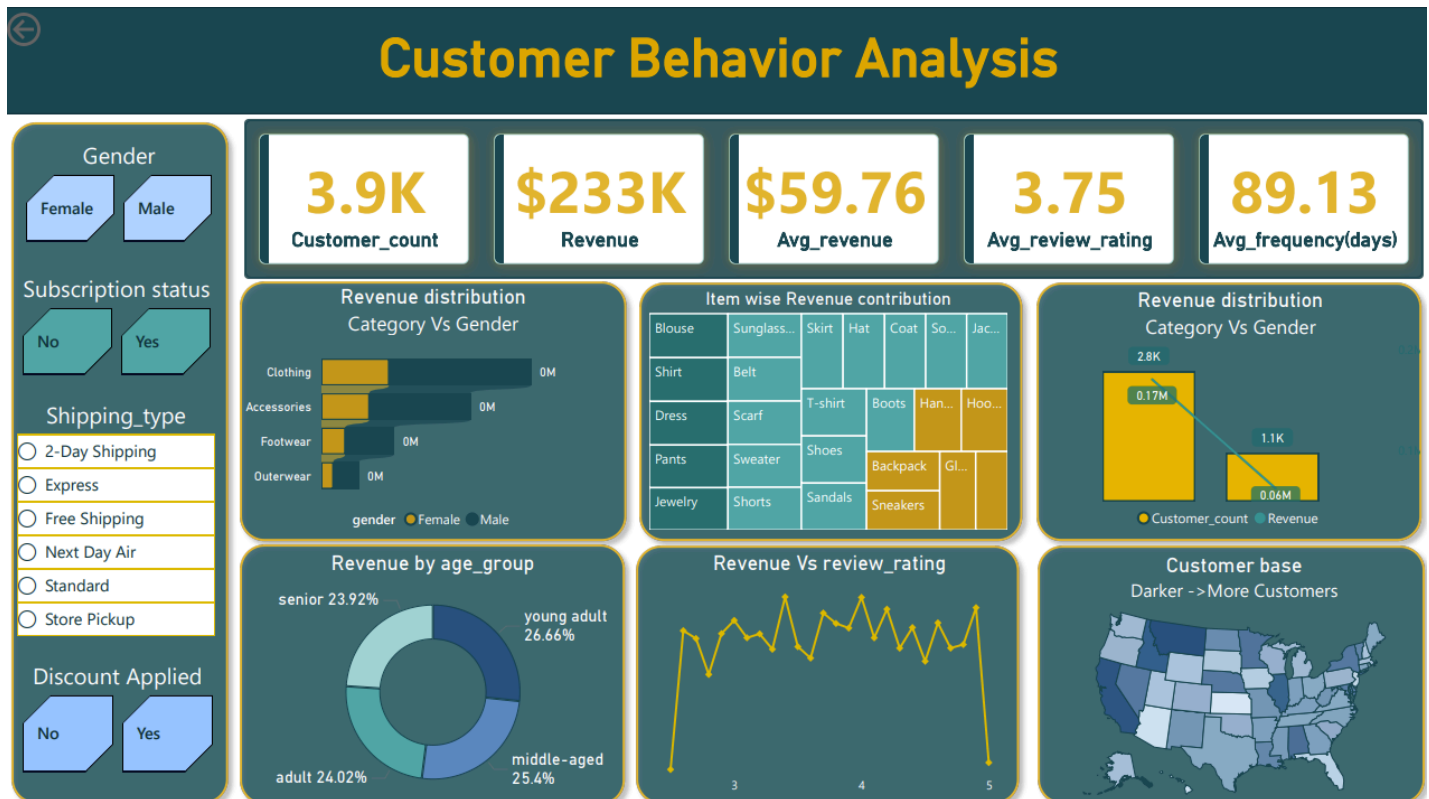
8.5 Verify Database structure: After establishing the connection, schemas, tables, and data were reviewed in pgAdmin to confirm that the database was correctly configured.

Connecting PostgreSQL to Power BI

8.6 Connect Power BI to PostgreSQL: PostgreSQL database was connected to Power BI using the Get Data → PostgreSQL Database option and by entering the Supabase database credentials.

8.7 Import Data: Relevant tables and query results were imported into Power BI for visualization.

8.8 Create Visualizations and Dashboard: Various charts and visualizations were created to present key insights derived from the SQL analysis as below



Slicers Used

- Gender
- Subscription Status
- Shipping Type
- Discount Applied

KPI Cards

Five cards are displayed at the top summarising the overall data:

- Customer Count — 3.9K
- Total Revenue — \$233K
- Average Revenue per transaction — \$59.76
- Average Review Rating — 3.75
- Average Purchase Frequency — 89.13 days

Key Findings:

1. Revenue by Category vs Gender (Bar Chart) This bar chart compares revenue generated across four product categories, split by gender.

This bar chart compares revenue generated across four product categories, split by gender.

- Clothing leads with \$104K, followed by Accessories \$74K, Footwear \$36K, and Outerwear \$19K
- Male customers contribute higher revenue across all four categories
- Clothing and Accessories together account for nearly 76% of total revenue, making them the most critical categories for business growth

- **2. Item-wise Revenue Contribution (Treemap)** Blouse, Shirt, Dress, Pants, and Jewelry are the top revenue-generating products. Smaller tiles like gloves and Jeans contribute least.

3. Revenue by Age Group (Donut Chart) Revenue is almost equally contributed by all age groups in the order of Young Adult (26.66%), Middle-Aged (25.4%), Adult (24.02%), and Senior (23.92%).

4. Revenue vs Review Rating (Line Chart) Revenue fluctuates across ratings 3 to 5 with no strong linear trend, indicating customer spending is not heavily influenced by ratings alone.

5. Customer Base by Location (USA Map) Montana, California, Idaho, Illinois and Alabama are the top 5 states with high customer base.

9. Recommendations

1. Double Down on Top Performing States: Montana, California, Idaho, Illinois, and Alabama have the strongest customer base. Prioritise these states for targeted digital ads and seasonal campaigns since the cost of conversion is lower with an already familiar audience.

2. Penetrate Low-Density States: States with low customer presence represent untapped potential. Introductory discounts, free shipping, and brand awareness campaigns in these regions can help acquire new customers.

3. Invest in Clothing and Accessories: Clothing (\$104K) and Accessories (\$74K) contribute 76% of total revenue. These categories should be prioritised for inventory planning, new launches, and seasonal promotions especially during peak seasons.

4. Revive Underperforming Products: Gloves and Jeans are the lowest contributors. Bundle them with best-sellers (e.g. Jeans + Belt), offer limited-time discounts, or consider reducing stock if sales do not improve after promotion.

5. Strengthen Male Customer Engagement: Male customers contribute higher revenue across all categories. A dedicated male-focused strategy — personalised recommendations and targeted email campaigns — can further boost revenue.

6. Improve Average Review Rating: The average rating of 3.75 out of 5 indicates moderate satisfaction. Identifying and fixing low-rated products for quality or description issues can push the rating above 4.0 — a key online trust benchmark.

7. Reduce Purchase Frequency Gap: Customers return every 89 days on average. A loyalty programme, re-engagement emails at the 60-day mark, and flash sales can shorten this cycle and increase repeat purchases.

8. Age-Specific Product Personalisation: All age groups contribute equally to revenue. Personalising recommendations by segment — trendy items for Young Adults, Outerwear for Seniors — can increase average order value across all groups.